06-21 第1篇

阅读理解（说明文）

① **Why Is the Sky Blue?**

② When you look up at the sky on a clear day, you see a beautiful blue color. But have you ever wondered why the sky is blue and not red, green, or yellow? The answer lies in how sunlight travels through the air around us.

③ Sunlight looks white, but it is actually made up of many different colors — like the colors you see in a rainbow. Each color has its own wavelength (波长). Blue light has a very short wavelength, while red light has a longer one. When sunlight passes through the earth's atmosphere, it meets tiny molecules (分子) of air. These molecules scatter (散射) the light in different directions.

④ Here is the key: short wavelengths like blue are scattered much more than long wavelengths like red. So when we look at the sky, we see mostly blue light that has been scattered toward our eyes from all around. The red and orange light travels more straight through and reaches us mainly when the sun is low — that is why sunsets often look red or orange.

⑤ The same idea explains why the sky looks darker blue when you are high up in the mountains. There is less air above you, so less scattering happens. Scientists call this effect "Rayleigh scattering" (瑞利散射), named after a British scientist who studied it more than a hundred years ago. Next time you enjoy a blue sky, you can remember the science behind it!

注释： wavelength 波长；molecule 分子；scatter 散射

选择题

1. According to the passage, what does sunlight look like but actually contain? A. Many different colors, like those in a rainbow. B. Only blue light. C. Only red and orange light. D. No color at all.

2. Why does the sky appear blue? A. Because the air itself is blue. B. Because blue light has a short wavelength and is scattered more than red light. C. Because the sun only sends blue light. D. Because our eyes see blue better than other colors.

3. Why do sunsets often look red or orange? A. Because the sun is hotter at sunset. B. Because blue light disappears at night. C. Because red and orange light travel

more straight and reach us when the sun is low. D. Because the sky changes color every evening.

4. What can we infer about the sky when you are high in the mountains? A. It looks the same as at sea level. B. It becomes red or orange. C. It has no color. D. It looks darker blue because there is less air above and less scattering.

5. What is the main purpose of this passage? A. To explain why the sky is blue using the science of light scattering. B. To tell a story about a scientist. C. To compare mountains and the sea. D. To teach readers how to paint the sky.

详细解析

1. 【答案】A

【题型】 细节理解

【定位】 第③段 "Sunlight looks white, but it is actually made up of many different colors — like the colors you see in a rainbow."

【解析】 原文明确说明阳光看起来是白色，但实际由多种颜色组成，像彩虹中的颜色。A 项正确。B 项仅为蓝色、C 项仅为红橙、D 项无颜色均与原文矛盾。故选 A。

2. 【答案】B

【题型】 细节理解

【定位】 第④段 "short wavelengths like blue are scattered much more than long wavelengths like red. So when we look at the sky, we see mostly blue light that has been scattered toward our eyes"

【解析】 短文指出蓝光波长短、被散射更强，因此我们看到的天空主要是被散射的蓝光。B 项正确。A 项空气本身为蓝色、C 项太阳只发光、D 项眼睛更易见蓝均无依据。故选 B。

3. 【答案】C

【题型】 细节理解 / 推理判断

【定位】 第④段 "The red and orange light travels more straight through and reaches us mainly when the sun is low — that is why sunsets often look red or orange."

【解析】 红橙光波长较长，穿透更直接，当太阳偏低时更多到达我们，故日落常呈红橙色。C 项正确。A 项太阳更热、B 项蓝光消失、D 项每天变色均与原文不符。故选 C。

4. 【答案】D

【题型】 推理判断

【定位】 第⑤段 "The sky looks darker blue when you are high up in the mountains. There is less air above you, so less scattering happens."

【解析】 高山处上空空气少，散射减少，天空呈现更深的蓝色。D 项正确。A 项与海拔相同、B 项变为红橙、C 项无色均与原文推断矛盾。故选 D。

5. 【答案】A

【题型】 主旨理解 / 写作意图

【定位】 全文标题 "Why Is the Sky Blue?" 及第②段 "The answer lies in how sunlight travels through the air around us."

【解析】 本文旨在用光的散射原理解释天空为何呈蓝色。A 项准确概括。B 项讲科学家故事、C 项比较山与海、D 项教画天空均非主旨。故选 A。